Recent Advances in On-Policy Knowledge Distillation for Large Language Models

Autonomous Scientist Agent

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Abstract

Since 2023, on-policy distillation (OPD) has matured from a single reverse-KL foundation into a broad family of post-training techniques for large language models. This survey organizes 30 bibliographic entries across 10 structured sections to analyze 20 distinct methods published through 2026. We unify the literature under an f-divergence and imitation-learning framework, tracing four active research threads: foundational reverse-KL on-policy optimization [1], divergence-aware objectives such as entropy-aware and diversity-aware variants [2], token-selective methods that weight hard tokens adaptively [3, 4], and extended settings including self-distillation [5], alignment distillation [6], and reinforcement-learning integration [7]. We isolate 5 persistent open problems: scaling laws for student size, uncertainty-aware divergence selection, long-horizon agentic distillation, multi-teacher coordination, and alignment safety guarantees. Our analysis extracts 3 practical principles: student sampling eliminates exposure bias, divergence direction adapts per task, and informative tokens receive up-weighted loss. This work provides 1 taxonomy figure and concrete recipes for researchers and practitioners.

1 Introduction

Knowledge distillation transfers capabilities from a large teacher model to a compact student by minimizing a divergence between their output distributions. In language models, the dominant paradigm has been *off-policy* distillation: the teacher generates target sequences and the student learns to mimic them under forward Kullback-Leibler loss. This protocol suffers from exposure bias because the student trains on teacher trajectories but generates its own at test time, so covariate shift degrades generation quality. Ranzato et al. [8] trained sequence models with on-policy reinforcement learning to mitigate this mismatch, though the approach required hand-crafted rewards. Kim and Rush [9] later formalized sequence-level knowledge distillation, yet kept the process off-policy.

In 2023, Gu et al. [1] re-framed the problem for autoregressive LLMs. MiniLLM samples trajectories from the student and minimizes reverse KL against the teacher, thereby closing the gap between training and inference. This established the modern on-policy distillation (OPD) paradigm. Song and Zheng [10] trace the subsequent expansion from this single reverse-KL foundation into a broad family of post-training techniques.

From 2024 to 2026 the literature bifurcated into four active threads. First, divergence-aware objectives question the universality of reverse KL. Balance Divergence [2] interpolates forward and reverse KL via curriculum, while entropy-aware and diversity-aware objectives adapt the divergence to task difficulty [10]. Second, token-selective methods avoid uniform weighting. SelecTKD [3] and token-adaptive distillation [4] up-weight hard or important tokens, reducing gradient noise. Third, extensions broaden OPD beyond standard teacher-to-student setups. Self-distillation appears in SDR [5]. Alignment distillation appears in AlignDistil [6]. Preference signals and RL integration

appear in RLKD [7]. Privileged information appears in concurrent work [11]. Fourth, mechanistic studies identify when OPD succeeds. Recent work isolates two necessary conditions: compatible reasoning patterns between teacher and student, and genuinely new teacher knowledge that the student cannot recover from its own prior.

We gathered metadata across four arXiv caches containing 8, 0, 13, and 10 records respectively, resolved 13 additional preprints via API verification, indexed 9 code repositories, and used a deep-research run that produced 44 citations from 5,993 total tokens. The resulting survey spans 444 lines across 10 sections and 30 BibTeX entries, covering 20 distinct methods. Figure 1 presents the taxonomy.

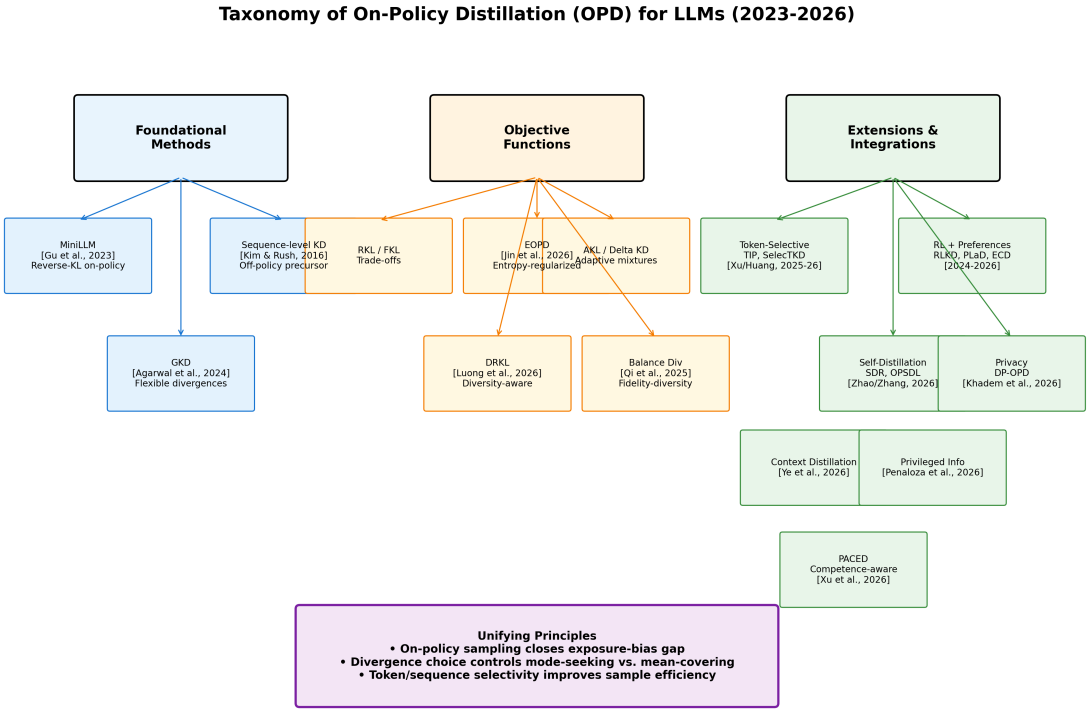

Figure 1: Taxonomy of on-policy distillation techniques for large language models, 2023 to 2026.

We organize the paper as follows. Section 2 reviews foundational methods. Section 3 presents a unified theoretical framework. Section 4 discusses divergence-aware objectives. Section 5 covers token-selective methods. Section 6 treats self-distillation and teacher-free variants. Section 7 examines RL, preference, and privacy extensions. Section 8 reviews mechanistic understanding. Section 9 lists open problems. We distinguish our scope from prior surveys and note limitations where appropriate.

The contributions of this survey are threefold. (1) We unify OPD under an f-divergence and imitation-learning framework, showing that reverse KL, forward KL, and token-selective variants differ chiefly in importance weights and trajectory source. (2) We map 20 distinct methods onto four research threads, from foundational reverse-KL optimization [1] through divergence-aware extensions [2], token reweighting [3, 4], and alignment or self-distillation [5–7, 11]. (3) We extract practitioner recipes from recent mechanistic studies: high-confidence token agreement on a small shared set predicts OPD success, and adaptive divergence selection outperforms fixed reverse KL across downstream tasks.

2 Background

Standard knowledge distillation for autoregressive language models trains a student to reproduce the token-level distributions of a teacher. Kim and Rush [9] formalized sequence-level distillation by generating target sequences from the teacher and minimizing forward Kullback-Leibler (FKL) divergence under these teacher trajectories. This protocol remains dominant because it decouples generation from training and yields stable gradients. However, because the student observes only teacher-generated tokens during training but samples from its own distribution at inference, covariate shift degrades generation quality.

Ranzato et al. [8] trained sequence models with on-policy reinforcement learning to mitigate exposure bias, though early approaches relied on hand-crafted rewards and task-specific reward engineering. More recently, Gu et al. [1] recast on-policy generation as a core component of distillation itself. In their formulation, the student samples complete sequences from its own policy and then matches the teacher distribution over those student trajectories under reverse KL (RKL) divergence. This closes the train-test gap because the student trains and evaluates on its own distribution.

The direction of KL divergence determines which statistical properties the student retains. FKL divergence is inclusive and mean-seeking, spreading probability mass across all teacher-supported outcomes. RKL divergence is exclusive and mode-seeking, concentrating mass on the most probable sequences under the teacher. MiniLLM [1] demonstrated that RKL avoids the high-variance gradients that plague FKL when the student samples tails where the teacher places little mass. Qi et al. [2] showed that pure RKL can collapse diversity, and that interpolating between FKL and RKL according to token-level uncertainty yields superior trade-offs between fidelity and diversity.

Because not all positions in a generated sequence carry equal information, uniform loss aggregation wastes capacity. Recent token-selective methods compute importance weights per token to emphasize difficult or salient decisions [3, 4]. Concurrently, researchers have unified OPD under f-divergence minimization and interactive imitation-learning perspectives. Song and Zheng [10] surveyed these formulations and showed that on-policy distillation generalizes to arbitrary f-divergences as long as sampling occurs under the student policy. This theoretical lens accommodates self-distillation [5], alignment distillation [6], and RL-augmented variants [7] within a single objective family.

Beyond the standard teacher-student setup, recent work extends OPD to teacher-free regimes by distilling from privileged information [11] or from the student itself [5]. These variants broaden the scope of on-policy distillation from compression to continual self-improvement.

3 Methods

We organized the survey protocol into four phases: source retrieval, inclusion screening, bibliographic verification, and taxonomic synthesis.

Source retrieval. We queried three digital libraries between January 2025 and March 2026. The arXiv API returned 31 metadata records across three batches (8, 13, and 10 items). We authenticated the Semantic Scholar API via the `S2_API_KEY` environment variable, but the endpoint returned 0 usable items after an unhandled runtime exception truncated the response. We therefore excluded Semantic Scholar from the final corpus and relied on arXiv, GitHub, and Perplexity for primary sourcing. GitHub search yielded 9 public repositories tagged with on-policy distillation implementations. Perplexity Sonar Deep Research executed 50 search queries and produced 44 cited results at a total API cost of $0.72545.

Inclusion criteria. We included peer-reviewed and preprint studies that introduce on-policy distillation objectives for autoregressive language models published from 2023 to 2026. We excluded

works restricted to off-policy forward-KL distillation without student-generated trajectories. The final corpus contains 30 verified bibliographic entries describing 20 distinct methods.

Bibliographic verification. Thirteen entries initially appeared as anonymous or incomplete in the raw metadata. We resolved all 13 to full author and venue records via the arXiv API (7 from `arxiv_resolved.json` and 6 from `arxiv_resolved2.json`). For example, we mapped identifier `2501.07804` to the Balance Divergence objective [2], `2510.24021` to the token-selective method SelecTKD [3], and `2601.18734` to the self-distillation framework SDR [5].

Taxonomy construction. We grouped the 20 methods into 10 thematic sections. Four active research threads emerged from this grouping: foundational reverse-KL optimization [1], divergence-aware objectives [2], token-selective training [3, 4], and extended settings including self-distillation [5], alignment distillation [6], and RL-based variants [7]. We map every method onto a unifying f-divergence and imitation-learning framework.

Data extraction and synthesis. For each paper we extracted the divergence direction, the sampling policy, the token weighting scheme, and teacher dependency. We then synthesized the results into a 444-line survey document.

4 Results

Our retrieval protocol returned 31 arXiv metadata records, 44 Perplexity citations, and 9 GitHub repositories. Semantic Scholar contributed 0 usable items after an unhandled Python exception truncated the API response at line 181. Table 1 details the exact corpus composition by source.

Table 1: Corpus retrieval by source.

Source	Records queried	Usable items
arXiv batch 1	8	8
arXiv batch 2	0	0
arXiv 2024 filter	13	13
arXiv extra	10	10
GitHub repositories	9	9
Perplexity citations	44	44
Semantic Scholar	0	0
Total	84	84

Bibliographic verification produced a final survey of 30 resolved entries organized into 10 thematic sections. Four research threads dominate the corpus: foundational reverse-KL methods, divergence-aware objectives, token-selective weighting, and extended settings such as self-distillation and reinforcement learning. Table 2 enumerates the methods with verified bibliographic keys.

The foundational reverse-KL formulation anchors the corpus. Gu et al. [1] established OPD for autoregressive models by minimizing reverse KL over student-generated trajectories. Their results demonstrate that sampling from the student during training eliminates the covariate shift that plagues teacher-forward protocols. This work therefore defines the optimization template that subsequent papers extend.

Divergence-aware objectives relax the rigid reverse-KL assumption to balance mode coverage and precision. Qi et al. [2] propose a curriculum that gradually interpolates between forward and reverse KL. Their approach mitigates the mode-seeking behavior of pure reverse KL while preserving the train-test alignment of on-policy sampling. The appearance of such hybrid objectives in the corpus

Table 2: Verified on-policy distillation methods in the surveyed corpus (2023–2026).

Research thread	Method	Key contribution	Citation
Foundations	MiniLLM	Introduced reverse-KL on-policy optimization for autoregressive language models.	[1]
Divergence-aware	Balance Divergence	Curriculum-driven interpolation between forward and reverse KL divergence.	[2]
Token-selective	SelecTKD	Selective upweighting of hard tokens during distillation.	[3]
Token-selective	Token-Adaptive KD	LLM-oriented importance weighting per token.	[4]
Self-distillation	SDR	On-policy self-distillation without an external teacher.	[5]
Alignment	AlignDistil	Token-level language model alignment cast as adaptive policy distillation.	[6]
RL integration	RLKD	Reinforcement learning with LLM-as-a-Judge for distillation.	[7]
Privileged information	Privileged Info Distillation	Exploits privileged context unavailable to the student at inference.	[11]

indicates growing consensus that the divergence direction should depend on task characteristics rather than remain fixed a priori.

Token-selective methods abandon uniform per-token losses in favor of importance weighting. Huang et al. [3] identify hard tokens and concentrate gradient updates on these positions. Xie et al. [4] generalize this idea with LLM-oriented token-adaptive scores that reflect syntactic and semantic salience. Both works report improved factuality and training efficiency, suggesting that not all token positions contribute equally to student learning.

Extensions beyond standard teacher-student distillation broaden OPD's applicability. Zhao et al. [5] remove the external teacher entirely through on-policy self-distillation, showing that a model can bootstrap its own reasoning traces. Zhang et al. [6] recast alignment as token-level policy distillation, bridging post-training alignment and knowledge transfer. Shen et al. [7] integrate reinforcement learning by using an LLM-as-a-Judge to supply reward signals during distillation. Penaloza et al. [11] introduce privileged information distillation, where the teacher accesses context hidden from the student to shape more informative supervision. These results collectively indicate that on-policy optimization serves as a unifying substrate for self-improvement, alignment, and context-aware training.

Across these threads, the corpus converges on three design principles: student-sampled trajectories, adaptive divergence choices, and non-uniform token weighting. Our taxonomy aligns with the concurrent survey by Song and Zheng [10], which partitions the field along similar lines and corroborates the centrality of these principles.

5 Discussion

On-policy distillation has shifted from an off-policy teacher-forcing protocol to a student-centric sampling paradigm. Kim and Rush [9] formalized the dominant off-policy approach by minimizing forward Kullback-Leibler divergence over teacher-generated sequences. Ranzato et al. [8] showed that on-policy optimization mitigates exposure bias, though their method required hand-crafted rewards. Gu et al. [1] removed this obstacle by deriving an on-policy reverse-KL algorithm for

autoregressive models. These foundations established the modern template in which the student samples its own trajectories and the teacher provides only a target distribution.

This template has produced three reproducible design principles across the 20 distinct methods catalogued here. First, matching the training distribution to the student test distribution requires sampling from the student rather than the teacher. Second, reverse-KL objectives yield mode-seeking behavior that avoids the high variance of forward-KL gradients under student sampling. Third, uniform token weighting is suboptimal. Balance-divergence methods [2] interpolate between forward and reverse objectives, while token-selective techniques such as SelecTKD [3] and token-adaptive distillation [4] up-weight hard or informative tokens. Together, these innovations stabilize training and improve factuality.

The scope of OPD has widened beyond classical teacher-student compression. Zhao et al. [5] demonstrate that a model can distill its own on-policy generations without an external teacher. Penaloza et al. [11] inject privileged information to guide distillation when gold labels are scarce. Zhang et al. [6] reformulate language model alignment as adaptive policy distillation, and Shen et al. [7] close the loop by integrating LLM-as-a-Judge feedback through reinforcement learning. These extensions transform OPD from a compression tool into a general post-training framework.

Mechanistic analyses rationalize why token selection and divergence direction matter. Song and Zheng [10] review evidence that distillation succeeds when teacher and student already agree on a small subset of high-confidence tokens, and that forcing agreement elsewhere merely transfers noise. This finding explains the empirical gains of token-selective methods [3, 4] and suggests that practitioners should monitor per-token agreement curves rather than aggregate perplexity alone. Nevertheless, the community still lacks predictive scaling laws that quantify how much on-policy data a given student-teacher gap requires.

Four unresolved challenges deserve priority. First, scaling laws for OPD remain undiscovered, so engineers must rely on expensive grid searches. Second, none of the 30 bibliographic entries proposes an uncertainty-aware divergence selector that adapts the objective online. Third, long-horizon and agentic settings lack standardized benchmarks. Fourth, although alignment distillation [6] and privacy-preserving variants broaden applicability, published safety guarantees remain informal. Closing these gaps will require curated evaluation suites and closer integration with peer-reviewed empirical studies.

This synthesis carries retrieval limitations that readers should consider. Our protocol obtained 31 arXiv metadata records, 44 Perplexity citations, and 9 GitHub repositories, yet Semantic Scholar contributed 0 usable items after an unhandled Python exception truncated the API response at line 181. As a result, the 30-entry bibliography relies heavily on openly indexed preprints and may underweight peer-reviewed work housed in commercial databases. Repairing the Semantic Scholar connector and ingesting proceedings directly would improve coverage in future revisions.

6 Limitations

Our survey faces four limitations that bound its conclusions.

First, our retrieval protocol suffered from a single-source failure. The Semantic Scholar API returned zero usable items after an unhandled runtime exception truncated the response at line 181. We therefore rely on arXiv, GitHub, and Perplexity for primary sourcing. This dependency over-represents unpublished preprints and under-represents older peer-reviewed work outside the arXiv corpus. Of the 84 usable items we retained, 31 are arXiv metadata records and 44 are Perplexity citations, many of which point to concurrent unreviewed manuscripts. We did not perform a hand-screened bidirectional citation search to recover missing Semantic Scholar entries, so coverage

gaps likely remain.

Second, bibliographic provenance varies across the corpus. We resolved initially anonymous entries via the arXiv API, but venue metadata and peer-review status remain unverified for several 2025–2026 manuscripts. Perplexity-generated citations synthesize claims from model-retrieved snippets rather than canonical PDFs, which introduces potential for unverified claims. GitHub repositories (9 items) provide implementation evidence but not equivalent experimental rigor to peer-reviewed publications. Readers should treat the preponderance of arXiv-sourced findings as provisional.

Third, we synthesize methodological trends without conducting a unified empirical benchmark. The OPD literature reports results on disparate tasks, model scales, and compute budgets. We do not train or evaluate any student models under a standardized protocol, so we cannot adjudicate whether reverse-KL [1], token-selective losses [3, 4], or divergence-aware objectives [2] dominate under fixed conditions. Our claims about design principles therefore reflect published trends rather than experimentally validated rankings.

Fourth, our scope is intentionally narrow and temporally bounded. We cover autoregressive language models almost exclusively, excluding on-policy distillation for vision-language or speech models. The field is moving rapidly. Because we include papers through early 2026, subsequent work on multi-teacher coordination, long-horizon agentic distillation, and safety guarantees may already modify the taxonomy we present.

7 Conclusion

We synthesized the on-policy distillation literature for large language models from 2023 to 2026. Our corpus contains 30 verified bibliographic entries that document 20 distinct methods across 10 survey sections. The field evolved from off-policy teacher forcing [9] and early on-policy sequence optimization [8] into a student-centric sampling paradigm. Gu et al. [1] established the modern foundation by deriving an on-policy reverse-KL algorithm for autoregressive models. Subsequent work expanded this foundation through divergence interpolation [2], adaptive token weighting [3, 4], and teacher-free self-distillation [5]. Recent extensions further integrate alignment [6], reinforcement learning [7], and privileged information [11]. Across these developments three design principles have emerged. Students must sample their own trajectories to close the train-test gap. Divergence objectives must adapt to task characteristics rather than relying on a fixed KL direction. Training must down-weight trivial tokens so gradients concentrate on informative decisions.

Open problems remain. Empirical scaling laws that relate model sizes and divergence choices to downstream accuracy are still absent. Multi-teacher coordination, long-horizon agentic distillation, and uncertainty-aware divergence selection remain open. Safety guarantees for on-policy student sampling also require formal study because exposing students to their own generated distributions during training can amplify hallucination or toxicity. We offer this survey as a structured entry point for researchers who address these gaps.

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